

DynamicDex: Contact-Aware Dexterous Grasping under Hidden Load Shifts

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Abstract—Hidden mass redistribution changes the support required from a dexterous grasp even when object geometry and total mass remain unchanged. We present DynamicDex, a residual control framework that adapts finger support from local tactile feedback and global wrist loading while limiting departures from an established grasp. A shared encoder conditions each finger’s features on wrist wrench, and a recurrent actor maps recent observations to bounded joint adjustments. An asymmetric critic uses privileged physical state during training; the actor executes from onboard observations. An integrated simulation and physical testbed enables controlled payload redistribution and tactile–wrench recording. With a fixed wrist, DynamicDex raises moving-load pose success from 80% to 100% and reduces mean squared normalized joint torque by approximately 40% relative to fixed-grasp control. Basket accounts for the success gain, improving from 12/30 to 30/30 successful trials. Physical Basket grasps achieve 90% success under static loads and 80% under moving loads, with mean contact retention of 95% and 85%, respectively. Contact-level analysis shows that successful pose recovery can coexist with interrupted finger support, distinguishing recovery of the object from continuity of the grasp. Project page: <https://anonymous.4open.science/r/ICRA2027-DynamicDex-12D6>.

I. INTRODUCTION

An established grasp can lose support when a payload moves inside an object. A basket with a shifting load or a pan carrying an off-center payload can retain the same visible shape and total mass while imposing a different moment on the hand. The hand must redistribute support, connecting grasp synthesis, contact-rich execution, and policy learning [1]–[3]. Object pose alone is insufficient to characterize this process: the object may remain upright as a supporting finger loses contact.

Tactile control enables force adaptation, finger gaiting, and regulation under changing loads [4]–[6]. CoordGrasp coordinates contacts and arm–hand motion during uncertain grasp execution [7], while Grasp-to-Act adapts functional tool grasps [8]. We study how bounded finger adjustments respond to hidden mass redistribution after grasp acquisition, using tactile observations together with wrist wrench.

These signals describe different aspects of the same interaction. Wrist wrench measures the resultant loading but does not identify which finger is losing support; local tactile observations describe individual contacts but do not determine the total external moment. Load-conditioned tactile representations [9] offer a way to relate the two. Additional joint freedom, however, can alter contact geometry as well as force. A controller must therefore regulate departures from its reference grasp, and its evaluation must measure both object motion and finger support.

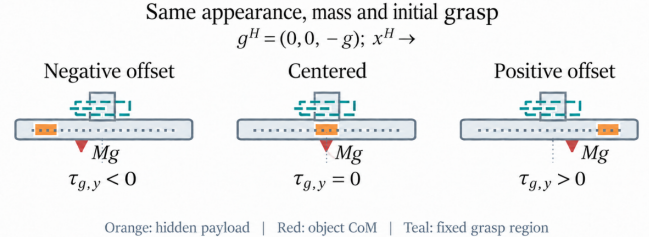


Fig. 1. Hidden payload redistribution at unchanged geometry, total mass, and initial grasp. Moving the payload (orange) shifts the object CoM (red) relative to the grasp region (teal), changing the gravitational moment while total weight Mg remains constant.

DynamicDex conditions shared finger features on wrist loading, integrates a short causal history, and outputs joint residuals subject to reference, joint, and rate bounds. Its actor uses tactile descriptors, proprioception, and wrist wrench; object pose and mass properties are available only to the training critic. We evaluate adaptation by varying payload position while keeping exterior geometry and total mass fixed (Fig. 1), following controlled-loading studies of human grasp coordination [10]. Static tests vary the load distribution between grasps, and moving-load tests perturb it during holding. Fixed- and admittance-wrist comparisons distinguish finger adaptation from wrist compliance.

With a fixed wrist, DynamicDex raises moving-load pose success by 20 percentage points over fixed-grasp control and lowers mean squared normalized joint torque by approximately 40%. Basket accounts for all additional successes (30/30 versus 12/30). Contact measurements also reveal interruptions in finger support during successful pose recovery. The UR5e–LEAP testbed with four GelSight Mini sensors also supports physical Basket trials.

Our contributions are: (1) a load-conditioned residual controller combining global loading with local contact history; (2) a complete simulation and physical experimental platform integrating robot and sensor models, movable payloads, and tactile–wrench recording; (3) matched static-load, moving-load, and wrist-mode comparisons of action spaces and sensing representations; and (4) an analysis relating object-level recovery to finger-level support and physical load responses. The simulated platform includes four object scenarios; quantitative policy evaluation uses three, and physical experiments focus on Basket.

II. RELATED WORK

A. Dexterous Grasping and Contact Adaptation

Grasp synthesis identifies configurations that support an intended task. Task-oriented synthesis matches achievable and required wrench spaces [11]; BODex and Dexonomy generate grasps through bilevel optimization and a grasp taxonomy, respectively [12], [13]. DemoGrasp adapts a single grasp demonstration [14], while AdaDexGrasp combines geometry with finger-associated tactile observations for refinement [15].

During execution, contact control responds to changing interaction forces. HFTS combines force adaptation with finger gaiting [4], and shear-based control adapts to changes in weight and center of mass [5]. TacDexGrasp constrains contact-force ratios to resist rotational slip [6]; CoorGrasp coordinates contact forces and arm–hand motion [7]. Wrench-aware admittance addresses unknown payloads during transport [16]. DynamicDex studies bounded adaptation around an acquired grasp, with matched finger-action and wrist-mode comparisons under hidden load shifts.

B. Tactile–Wrench Representations and Policy Learning

Tactile representations range from force estimates in FEATS and FeelAnyForce [17], [18] to cross-sensor features in Transferable Tactile Transformers [19] and mechanics-based simulation in TaCauchy [20]. TacForeSight conditions tactile dynamics on wrist wrench [9], while DeCAL fuses visual and tactile observations in a contact-aware latent space [21]. DexRepNet++ encodes hand–object geometry [22], and Dex-X incorporates simulated tactile information [23]. We use wrist loading to condition per-finger contact features and integrate their recent history for control.

Residual learning [24], functional grasp adjustment in Grasp-to-Act [8], and progressive kinematic–dynamic alignment [25] provide approaches to adapting reference behavior. Human-in-the-loop reinforcement learning supports physical skill acquisition [26]. DynamicDex learns bounded residuals in simulation and freezes the selected actor for evaluation. Its experiments examine how action dimension, sensing, history, and wrist motion affect pose recovery and contact continuity.

III. PROBLEM FORMULATION

We consider a 16-joint hand with nominal grasp q_{ref} , reference hand–object pose ${}^H T_O^*$, and a set of required finger contacts. The palm frame is H ; the geometric object frame O is independent of payload position. Controllers share grasp acquisition, actuator settings, and commanded wrist motion within each comparison. The object body moves freely under contact, allowing slip and contact migration as the internal payload shifts.

For total mass M , CoM c^H , and gravity g^H , the gravitational wrench about the palm origin is

$$w_g^H = \begin{bmatrix} Mg^H \\ c^H \times Mg^H \end{bmatrix}, \quad c^O = \frac{m_0 c_0^O + m_p r_p^O}{m_0 + m_p}. \quad (1)$$

Here m_0, c_0^O describe the object body, m_p, r_p^O describe the payload, and $M = m_0 + m_p$. With rotation R_{HO} and

translation p_{HO} from O to H , $c^H = R_{HO}c^O + p_{HO}$. At a fixed hand–object pose, payload displacement changes the gravitational moment by

$$\Delta \tau_g^H = m_p (R_{HO} \Delta r_p^O) \times g^H. \quad (2)$$

Equations (1)–(2) distinguish payload travel, whole-object CoM displacement, and object pose error. Payload motion also changes the combined inertia; low-acceleration holding emphasizes gravity while retaining transient inertial effects.

In an off-table, quasi-static grasp, contact and gravitational wrenches approximately balance:

$$\sum_i \left[p_i^H \times f_i^H + \ell_i^H \right] + w_g^H \simeq 0. \quad (3)$$

In Eq. (3), f_i^H acts on the object at p_i^H , measured from the palm origin, and ℓ_i^H is a contact-patch moment. The sum includes non-pad contacts. The controller aims to retain the nominal pose and required contacts within bounded actuation while supporting the external load.

IV. DYNAMICDEX

DynamicDex conditions finger features on wrist loading, integrates their history, and maps the actor output to bounded joint targets (Fig. 2). A privileged critic uses object state during training.

A. Load-Conditioned Contact History

The sensor-frame wrench is expressed about the palm origin using the sensor rotation R_{HF} and position p_{HF} in frame H :

$$f^H = R_{HF} f^F, \quad \tau^H = R_{HF} \tau^F + p_{HF} \times R_{HF} f^F. \quad (4)$$

After the transformation in Eq. (4), we subtract a baseline acquired after closure and before lift, then apply a causal first-order filter with a 0.04-s time constant. The resulting signal measures changes from the pre-lift load, with age and validity flags accompanying its six components. This baseline subtraction retains unmodeled hand and sensor contributions.

For finger i , the contact descriptor is

$$d_i = [b_i, \hat{N}_i, \hat{S}_{x,i}, \hat{S}_{y,i}, u_i, v_i, A_i, Q_i, \text{age}_i, \text{valid}_i]. \quad (5)$$

Equation (5) contains contact state b_i , estimated normal and signed shear loads $(\hat{N}_i, \hat{S}_{x,i}, \hat{S}_{y,i})$, centroid (u_i, v_i) , active area A_i , central-load fraction Q_i , and timing metadata. Simulation obtains these quantities from visible-region contact impulses. FEATS [17] supplies a force-estimation route for optical tactile sensors; descriptor aggregation and validity handling are part of our interface. The physical recordings in Sec. VII show optical marker displacement and do not constitute calibrated force estimates.

Each observation contains $q, \dot{q}, q_{\text{ref}}, a_{t-1}^{\text{exec}}$ (64 values), four descriptors (40), wrist wrench and metadata (8), unit gravity direction in the palm frame (3), and expected/protected roles (8):

$$o_t \in \mathbb{R}^{123}, \quad O_t = o_{t-7:t} \in \mathbb{R}^{8 \times 123}. \quad (6)$$

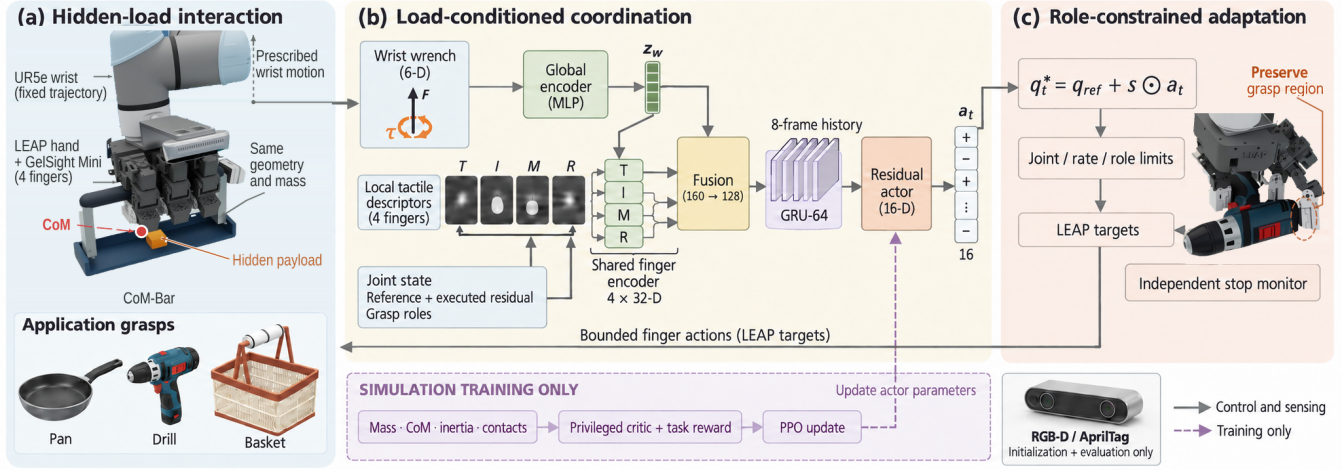


Fig. 2. Overview of DynamicDex. (a) Tactile descriptors, joint state, and wrist wrench describe grasping under hidden load shifts. (b) Wrist-conditioned finger embeddings are fused and processed over an eight-frame history to produce 16 joint residuals. (c) Reference, joint, and rate bounds map residuals to executable targets; role criteria enter the reward and monitor. Dashed paths denote training with a privileged critic. The camera supports initialization and evaluation.

Only samples available by the control tick enter Eq. (6). Object pose, mass, CoM, images, and payload labels are excluded.

A global MLP (32, 32) maps the 11 wrist/gravity inputs to $z_w \in \mathbb{R}^{32}$. Each finger contributes 32 local values: 16 joint-related values, ten tactile features, two role indicators, and a four-value identity code. Concatenating these with z_w feeds a shared MLP (64, 32). Four finger embeddings, ordered Thumb/Index/Middle/Ring, and z_w form 160 values. A 128-unit fusion layer, GRU-64, and 64-unit actor head output 16 Gaussian means with learned log standard deviations. The eight-frame history is sampled at 24 Hz. The GRU resets for each window and ignores padded frames.

B. Bounded Grasp Residuals

For sampled latent u_t , the actor proposes

$$a_t = \tanh u_t, \quad q_t^* = q_{\text{ref}} + s \odot a_t. \quad (7)$$

All joints use $s_j = 0.08$ rad for the three evaluated objects. Using the previous command $q^- = q_{\text{cmd}, t-1}$, we bound the proposal in Eq. (7) as follows:

$$\begin{aligned} L_j &= \max(q_{\min, j} + \epsilon_q, q_{\text{ref}, j} - s_j, q_j^- - v_{\max, j} \Delta t), \\ U_j &= \min(q_{\max, j} - \epsilon_q, q_{\text{ref}, j} + s_j, q_j^- + v_{\max, j} \Delta t), \\ q_{\text{cmd}, t} &= \text{clip}(q_t^*, L, U). \end{aligned} \quad (8)$$

Here ϵ_q is a joint-limit margin, $v_{\max, j}$ the command-rate limit, and Δt the control interval. An empty interval triggers a fault. The executed residual stored in the history is $a_{t, j}^{\text{exec}} = (q_{\text{cmd}, t, j} - q_{\text{ref}, j})/s_j$, or zero for $s_j = 0$. Each proposal remains referenced to q_{ref} .

For Grip-4D, $B_4 \in \mathbb{R}^{16 \times 4}$ contains one fixed closing direction per finger, each normalized by its largest absolute entry. Grip-4D sets $a_t = B_4 g_t$, $g_t \in [-1, 1]^4$, while Full-16 selects all joint components independently. Both use Eqs. (7)–(8); clipping can move the executed residual outside Grip-4D’s candidate span.

Equation (8) bounds joint commands; Cartesian contacts may still move. Roles enter the reward and contact monitor. Contact and role violations are secondary outcomes, not automatic pose failures. Basket retention requires three main fingers; its separate role criterion flags auxiliary-thumb loads above 0.2 N and enters training and recovery scoring. The camera and AprilTag serve initialization and evaluation.

C. Asymmetric Policy Training

The privileged critic observes mass, CoM, inertia, motion, contacts, friction, and remaining time, following asymmetric actor–critic learning [27]. Relative pose errors are

$$e_R = \|\text{Log}((R_{HO}^*)^T R_{HO})^\vee\|, \quad e_p = \|p_{HO} - p_{HO}^*\|. \quad (9)$$

The reward combines bounded pose, joint-torque, action-change, residual-magnitude, contact/role, and safety costs:

$$r_t = -\Delta t \sum_{k \in \{R, p, E, s, q, f, u\}} w_k c_k + b_s \mathbf{1}_{\text{success}} - b_f \mathbf{1}_{\text{failure}}. \quad (10)$$

Using the errors in Eq. (9), the pose costs in Eq. (10) saturate at 10° and 10 mm; joint torques are normalized by their respective limits. Specifically, c_E is the mean squared normalized joint torque, saturated per joint at one; $c_s = \|a_t^{\text{exec}} - a_{t-1}^{\text{exec}}\|^2/64$ and $c_q = \|a_t^{\text{exec}}\|^2/16$ penalize abrupt and large adjustments. The contact term combines the worst and mean squared required-finger deficits. The training configuration uses $w = (4, 2, 0.2, 0.05, 0.02, 1, 5)$. Dense contact costs penalize normal-load deficits below 0.5 N on required pads; safety costs increase above 85% of the joint-torque limit or 15 N pad force. The terminal coefficients are $b_s = b_f = 10$. Early failures also incur the discounted remaining-horizon failure cost.

Our PPO implementation [28] re-encodes observation windows and uses stored Gaussian latents for likelihood ratios. The critic is an MLP (256, 256, 128) with detached

Algorithm 1 DynamicDex training and execution

Require: Grasp references, load conditions, bounds

- 1: Initialize actor π_θ and privileged critic V_ψ
- 2: **while** training budget remains **do**
- 3: **for** each environment step in the rollout **do**
- 4: **if** a new episode is required **then**
- 5: Sample and configure the payload condition
- 6: Acquire grasp; record references
- 7: Build causal window O_t ; sample $u_t \sim \pi_\theta$
- 8: Execute projected targets from Eqs. (7)–(8)
- 9: Store residual, reward, and critic state
- 10: Compute advantages and update (θ, ψ) with PPO
- 11: Select the actor on validation trials and freeze θ
- 12: At evaluation, use $u_t = \mu_\theta(O_t)$ with the same bounds

TABLE I
OBJECT MASSES AND HIDDEN-LOAD RANGES.

Object	Total/payload [g]	Travel [mm]	CoM shift [mm]	Trials S/M
CoM-Bar	500/200	± 90	± 36	35/30
Pan	800/200	± 50	± 12.5	15/30
Basket	876/250	± 70	± 20	15/30

S/M: static/moving trials per controller. Mass is fixed within each trial.

actor-history features. Training uses 128-step rollouts and minibatches of 128 samples, four optimization epochs, learning rate 3×10^{-4} , discount 0.99, GAE 0.95, PPO clip 0.2, and entropy weight 0.003. Rollout boundaries preserve the physical grasp. Payload motion follows the sampled condition with fixed component masses. Evaluation uses the selected actor’s mean and the same bounds (Algorithm 1). We randomize friction following the general domain-randomization principle [29], over the range in Sec. V.

V. EXPERIMENTAL SETUP

We compare pose stability and finger support under static and moving hidden loads, varying action space, sensing, history, and wrist mode.

A. Objects and Evaluation Conditions

CoM-Bar, Pan, and Basket retain the same exterior geometry, total mass, and nominal grasp across load conditions (Table I). Figure 1 exposes the hidden payload for illustration. Drill uses an engineering controller in qualitative demonstrations only.

Static tests use seven equally spaced CoM-Bar positions and three positions for Pan and Basket, with five trials per position. Training samples payload locations continuously. Moving-load tests use two initial directions, three amplitudes, and five trials per setting. Component masses and central inertias remain fixed; combined inertia follows

$$I_C = \sum_k \left[R_k I_k R_k^T + m_k (\|d_k\|^2 I_3 - d_k d_k^T) \right], \quad (11)$$

where I_k is component k ’s central inertia, R_k its rotation into the common frame, d_k its offset from the combined

CoM, and I_3 the identity. Equation (11) accounts for payload displacement. Friction is sampled from $U(0.5, 0.8)$ and paired across controllers. Evaluation uses the trained geometries and load ranges.

The UR5e–LEAP simulation runs at 240-Hz physics and 24-Hz control. Formal trials close the hand on the supported object from registered arm and preclosure configurations; separate demonstrations approach from common Home. Each tactile grid has 24×32 cells, with contact hysteresis at 0.10/0.05 N. Retention requires all four fingers for CoM-Bar and Pan, and the three main fingers for Basket, excluding its auxiliary thumb.

B. Controllers and Training Budget

Fixed-grasp control holds the nominal grasp after common physical closure, without further preload optimization. Grip-4D and DynamicDex (Full-16) use the action spaces in Sec. IV with identical observations, history, reward, budget, references, and bounds.

TacDexGrasp-adapt. [6] combines tactile force estimates, friction-cone optimization, and joint tracking. Grasp-to-Act-adapt. [8] uses a 512-unit LSTM and measured-relative joint increments clipped to the common bounds. Both are platform adaptations using our sensing interface and load protocol.

The fixed wrist retains its nominal lifted pose. Simulation wrist admittance adjusts roll and pitch from torque changes relative to a stable loaded baseline, using a 0.15-s filter, gain 0.25 rad/(N·m), and per-axis angle/rate limits of 2.5° and $1.5^\circ/\text{s}$. Palm position remains fixed.

Each learned setting uses one 20,736-step training run, with validation every 6,912 steps. Selection prioritizes moving-load success, then static success and retention; the chosen actor is frozen for testing. Four retrained ablations remove tactile input, wrist input, history ($H = 1$), or conditioning in favor of same-input late fusion. Auxiliary features do not restore removed modalities.

The evaluation contains 325 static trials (five controllers, 65 conditions), 630 moving-load trials (seven controller/wrist combinations, 90 conditions), and 440 ablation trials (four variants, 50 static and 60 moving conditions): 1,395 complete tests across 20 groups. Ablations reuse the CoM-Bar/Pan reference trials without double counting. Trials are the units of comparison; results describe the frozen checkpoints and do not estimate variation across training runs.

C. Pose Success and Contact Retention

Static success requires at least 50 mm clearance, rotation within 10° , and translation within 10 mm throughout the 3-s hold, without a hard safety stop. Moving-load success applies these bounds throughout the final 2 s, allowing earlier deviations to recover. Initialization failures and safety stops remain in the denominator. The acquisition pose reference is retained after slip.

Required-contact retention is the fraction of scheduled hold samples with all required contacts present, averaged over trials. Unobserved samples after early termination contribute

TABLE II
CONTROLLER COMPARISON UNDER STATIC AND MOVING HIDDEN LOADS.

Controller	Static load						Moving load				
	CoM-Bar		Pan		Basket		Fixed wrist		Admittance wrist		T_{rec}
	Success	Ret.	Success	Ret.	Success	Ret.	Success	Ret.	Success	Ret.	[s]
Fixed grasp	34/35	71.9	12/15	13.3	15/15	100.0	72/90	66.7	75/90	66.7	1.00
Grip-4D	35/35	68.1	13/15	60.0	15/15	100.0	–	–	84/90	53.9	1.58
TacDexGrasp-adapt. [6]	35/35	100.0	11/15	100.0	8/15	100.0	–	–	30/90	96.5	1.53
Grasp-to-Act-adapt. [8]	35/35	9.5	0/15	48.1	0/15	100.0	–	–	30/90	57.1	1.70
DynamicDex	35/35	68.4	12/15	23.9	15/15	100.0	90/90	65.2	84/90	55.4	1.55

Success: passed trials/attempts. Ret.: failure-aware required-contact retention [%]. A dash denotes an unevaluated wrist mode. T_{rec} is the capped admittance-wrist recovery score.

zero. Contact, joint-torque, and role violations are recorded separately from pose success.

For observed lift/hold samples $\mathcal{V}$, mean squared normalized joint torque is

$$\overline{\tau_{\text{norm}}^2} = \frac{1}{16|\mathcal{V}|} \sum_{t \in \mathcal{V}} \sum_{j=1}^{16} \left(\frac{\hat{\tau}_j(t)}{\tau_{\text{lim},j}} \right)^2. \quad (12)$$

Equation (12) normalizes recorded torques $\hat{\tau}_j$ by joint limits $\tau_{\text{lim},j}$. Trial values are averaged and reported with observation coverage and safety stops. The independent monitor stops a trial at 25-N pad force, counting it as a failure.

The composite recovery score T_{rec} (Table II) requires 0.5 s of continuous valid execution after payload arrival, satisfying pose, clearance, contact, role, and torque criteria, with a 1% torque-limit tolerance. Unrecovered or censored arrivals receive 2 s. Scores are averaged over arrivals, then trials. This score assesses combined pose, contact, role, and torque recovery.

VI. SIMULATION RESULTS

A. Controller and Wrist-Mode Comparison

With a fixed wrist, DynamicDex raises moving-load pose success from 72/90 (80.0%) for fixed-grasp control to 90/90 (100.0%), a gain of 20 percentage points (Table II). All 18 additional successes occur on Basket (30/30 versus 12/30); both controllers pass all CoM-Bar and Pan trials. Mean squared normalized joint torque is approximately 40% lower (0.0215 versus 0.0358), with complete observation coverage and no safety stops in either group. Bounded finger adaptation therefore improves Basket recovery while lowering average normalized torque utilization across the evaluation set.

With admittance, DynamicDex passes 84/90 trials (93.3%) versus 75/90 (83.3%) for fixed grasp, retaining a 10-percentage-point advantage. Grip-4D matches 84/90, including the same six Pan failures. Admittance reduces DynamicDex's retention from 65.2% to 55.4% and adds six failures relative to its fixed-wrist setting. Wrist compliance provides no additional recovery benefit in these conditions; Grip-4D's fixed-wrist setting remains unevaluated.

Static success reaches 62/65 (95.4%) for DynamicDex, compared with 63/65 for Grip-4D and 61/65 for fixed grasp. DynamicDex and Grip-4D pass the same 35 CoM-Bar and 15 Basket trials, differing by one Pan success. Their similar

TABLE III
SENSING, HISTORY, AND FUSION ABLATIONS.

Variant	Static load				Moving load	
	CoM-Bar		Pan		CoM-Bar + Pan	
	Success	Ret.	Success	Ret.	Success	Ret.
No tactile	35/35	74.2	12/15	20.0	53/60	34.3
No wrench	35/35	76.0	12/15	60.0	55/60	34.1
No history ($H = 1$)	35/35	74.3	12/15	26.7	55/60	33.0
Late fusion	35/35	71.7	12/15	60.0	54/60	37.9
DynamicDex	35/35	68.4	12/15	23.9	54/60	33.1

Success: passed trials/attempts; Ret.: required-contact retention [%].
Moving-load tests use the same admittance wrist.

static performance contrasts with the larger gain over fixed fingers during load motion.

Under admittance, the platform-adapted TacDexGrasp and Grasp-to-Act controllers each pass 30/90 moving-load trials (33.3%), 60 percentage points below DynamicDex. Their static counts are 54/65 and 35/65. TacDexGrasp-adapt. retains required contacts for 96.5% of moving-load hold samples despite its lower success, showing that contact presence alone does not ensure pose stability.

DynamicDex's fixed-wrist retention is 65.2%, versus 66.7% for fixed grasp. On Pan, both pass all 30 trials despite simultaneous required-contact retention of just 7.0% and 0.0%, respectively. Under admittance, DynamicDex's composite recovery score is 1.55 s versus 1.00 s for fixed grasp, despite higher final pose success. The thumb-load role monitor flags 18/90 DynamicDex trials in each wrist mode. Improved pose recovery thus coexists with contact interruptions and violations of the evaluated role criterion.

B. Sensing and Representation Ablations

The CoM-Bar/Pan ablations show similar pose success within the fixed training budget (Table III). All variants pass the same 47/50 static trials. Moving-load success ranges from 53/60 without tactile input to 55/60 without wrist input or history; DynamicDex and late fusion both pass 54/60. Late fusion raises retention from 33.1% to 37.9%, a 4.8-percentage-point gain, without changing success. The full representation therefore provides no consistent advantage in success or retention on this subset. These admittance-wrist ablations exclude Basket, which accounts for the fixed-wrist

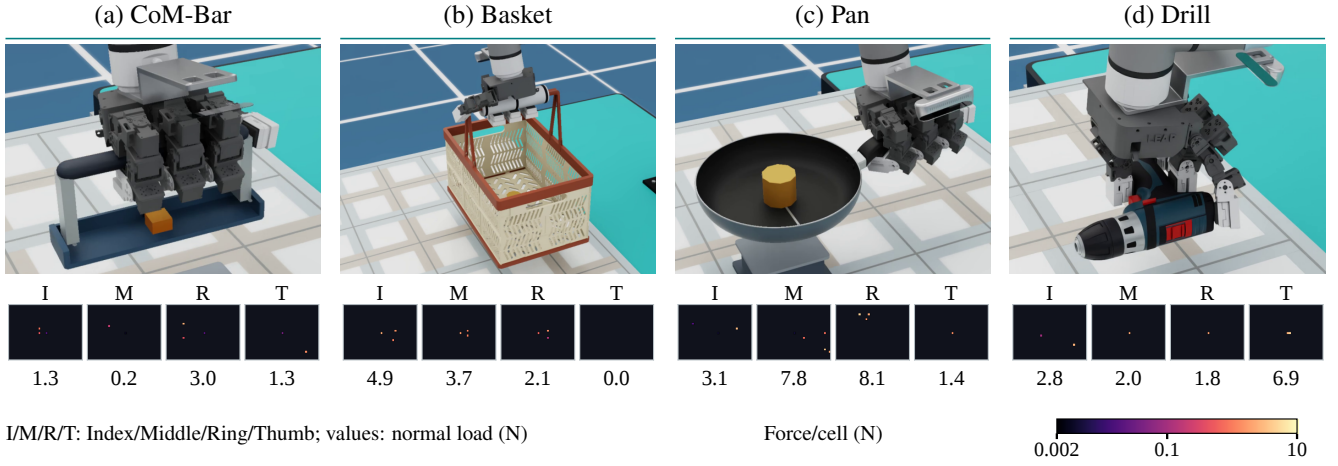


Fig. 3. Post-lift configurations and simulated tactile loads for (a) CoM-Bar, (b) Basket, (c) Pan, and (d) Drill. The first three use the frozen DynamicDex policy; Drill uses an engineering controller. Tactile tiles show per-cell normal force on a shared logarithmic scale, with per-finger totals in newtons below. These common-Home demonstrations are separate from the formal evaluation trials.

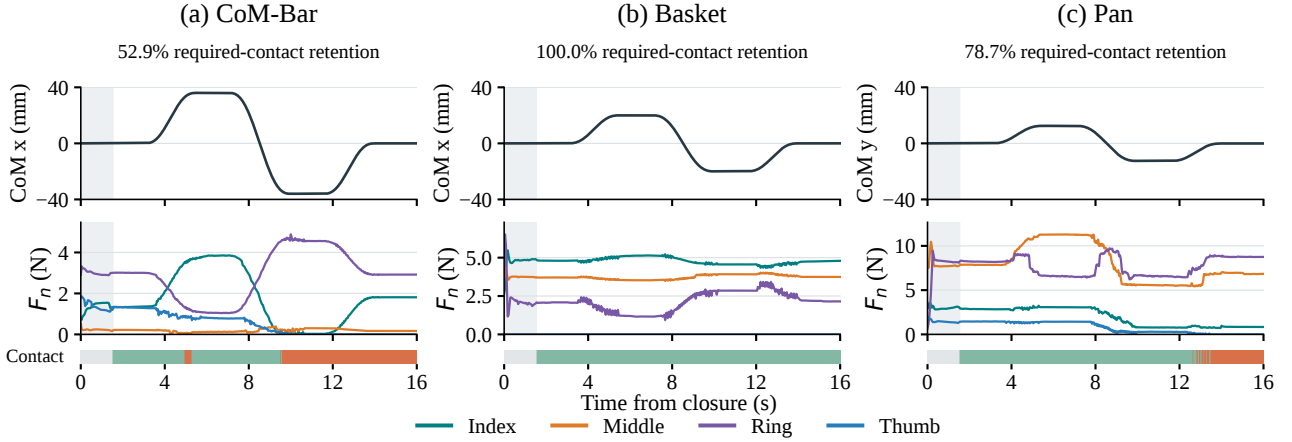


Fig. 4. Finger support during 16-s payload redistribution with wrist admittance: (a) CoM-Bar, (b) Basket, and (c) Pan. Rows show whole-object CoM displacement, finger normal loads, and required-contact presence. Green denotes retained support, orange a missing required contact, and gray the interval outside holding. Basket requires three primary fingers; CoM-Bar and Pan require four. All recover pose, but retention is 52.9%, 100%, and 78.7%, respectively, revealing contact interruptions in CoM-Bar and Pan.

success gain.

C. Observed Load Redistribution

Figure 3 shows the three frozen-policy grasps and the engineering-controlled Drill example. These common-Home demonstrations use scripted approach and policy activation after closure, separately from formal trials initialized at the nominal grasp. Tactile tiles quantify simulated per-cell normal force on a shared logarithmic scale and per-finger load totals; RGB and nearest tactile samples differ by 12.5 ms.

Figure 4 compares finger support in 16-s admittance recordings. CoM-Bar, Basket, and Pan have masses of 0.500, 0.876, and 0.800 kg and CoM excursions of approximately ± 36 , ± 20 , and ± 12.5 mm. The payload reaches its first side at 5.5 s, reverses by 10 s, and returns to center at 14 s. Unsmoothed 24-Hz curves and contact strips track the resulting support changes.

All three recordings recover object pose, but contact continuity differs. Basket retains its three primary contacts

throughout the hold; CoM-Bar and Pan achieve 52.9% and 78.7% retention, missing at least one required contact in 164/348 and 74/348 samples. Basket’s thumb remains unloaded, so its 100% retention refers to the primary fingers. The trajectories expose interrupted support that pose success alone misses.

In the matched Pan demonstrations (Fig. 5), fixing the wrist raises retention from 78.7% to 100% and reduces peak translation from 9.63 to 4.94 mm (48.7%) for the same actor. Peak rotation changes little, from 9.19° to 9.06° . Both recover within the $10^\circ/10$ -mm bounds using 0.10/0.05-N contact thresholds. The fixed wrist therefore improves contact continuity by 21.3 percentage points and nearly halves peak translation despite similar rotational stability.

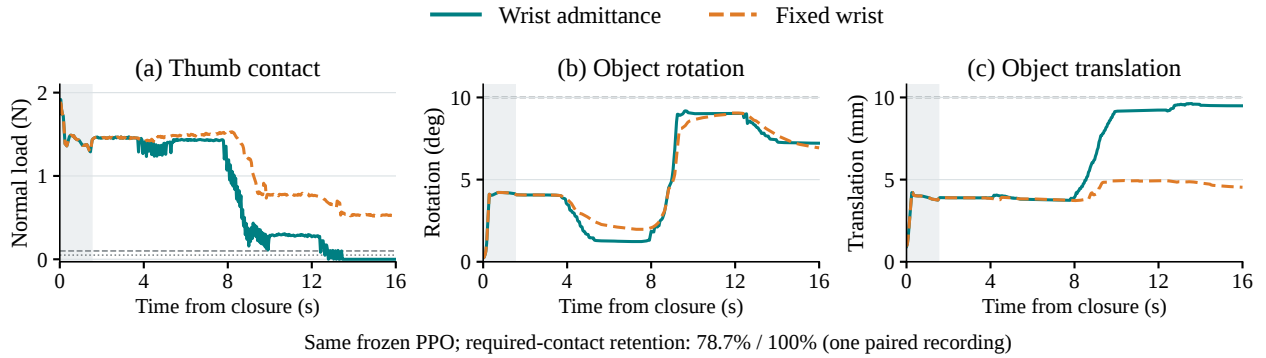


Fig. 5. One paired Pan recording with the same frozen actor under fixed (orange dashed) and admittance-controlled (teal solid) wrists. Panels show (a) thumb normal load, (b) object rotation, and (c) translation relative to the acquired grasp. Horizontal lines mark contact or pose thresholds. Both recover pose; fixing the wrist raises required-contact retention from 78.7% to 100% and reduces peak translation by 48.7% (9.63 to 4.94 mm).

TABLE IV
DYNAMICDEX (FULL-16) ON BASKET WITH WRIST ADMITTANCE.

Object	Static load ($N = 10$)			Moving load ($N = 5$)		
	SR	Ret.	$\bar{\theta}_{\max}$	SR	Ret.	$\bar{\theta}_{\max}$
Basket	9/10	95.0	4.2	4/5	85.0	8.6

SR: successes/attempts. Ret. [%] and $\bar{\theta}_{\max}$ [deg] are the means of trial-level contact retention and peak rotation, respectively.

VII. REAL-WORLD EXPERIMENTS

A. Platform and Basket Trials

The physical testbed integrates a UR5e, a 16-joint LEAP hand [30], four GelSight Mini sensors, and wrist wrench sensing. Timestamped tactile images and wrench streams record local contact deformation and global loading. Three main fingers grasp the foam-covered Basket handle, with the thumb providing an auxiliary contact.

We evaluate DynamicDex (Full-16) with wrist admittance in ten static-load and five moving-load Basket trials (Table IV). Static trials keep the payload position fixed during holding; moving trials shift it after the grasp is established. Contact retention and peak rotation are averaged over trials within each condition.

B. Grasp Stability under Load Changes

The physical grasp succeeds in 9/10 static-load trials (90%) and 4/5 moving-load trials (80%), with one failure in each condition. Static loading yields 95.0% mean required-contact retention and a mean peak rotation of 4.2°. Under moving loads, these values are 85.0% and 8.6°, respectively: retention decreases by 10 percentage points, and mean peak rotation increases by 4.4°. Moving loads therefore reduce contact continuity and increase angular excursions while retaining success in four of five grasps.

Figure 6 shows the accompanying changes in local contact deformation and wrist loading. Payload addition and displacement produce distinct marker-displacement patterns across the fingers. In the corresponding independent 10-s wrist recordings, the uncompensated mean tool-frame

torque about the y -axis changes from -0.620 N·m at initial grasp to -1.054 N·m after loading and -1.146 N·m after load displacement. The wrist traces distinguish the loading conditions, while the tactile images localize the associated contact deformation.

VIII. DISCUSSION AND CONCLUSION

DynamicDex combines wrist-conditioned tactile history with bounded joint residuals for hidden load shifts. Across moving-load simulations with a fixed wrist, it raises pose success from 80% to 100% and lowers mean squared normalized joint torque by approximately 40% relative to fixed grasp; Basket accounts for all additional successes. Under admittance, it retains a 10-percentage-point success advantage over fixed fingers and matches Grip-4D. CoM-Bar/Pan ablations show no consistent gain from the full representation within the training budget. Physical DynamicDex grasps with wrist admittance achieve 90% static-load and 80% moving-load success, with mean contact retention of 95% and 85%. Bounded finger adaptation improves pose recovery in these conditions; continuous support remains an unresolved control objective.

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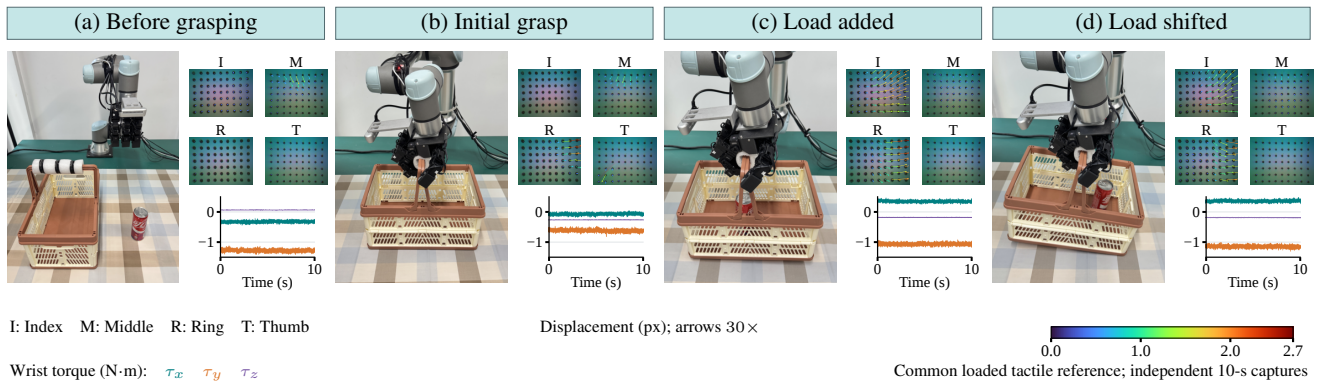


Fig. 6. Physical Basket loading conditions: (a) before grasping, (b) initial grasp, (c) payload addition, and (d) payload displacement. Tactile panels show marker motion relative to a common loaded reference; arrow lengths are magnified 30 $\times$, and color encodes displacement in pixels. Tool-frame wrist-torque traces are independent 10-s recordings matched to the conditions; the no-object reference was acquired after release. The observations show how loading changes wrist torque and local contact deformation. Marker displacement is not a calibrated force measurement.

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